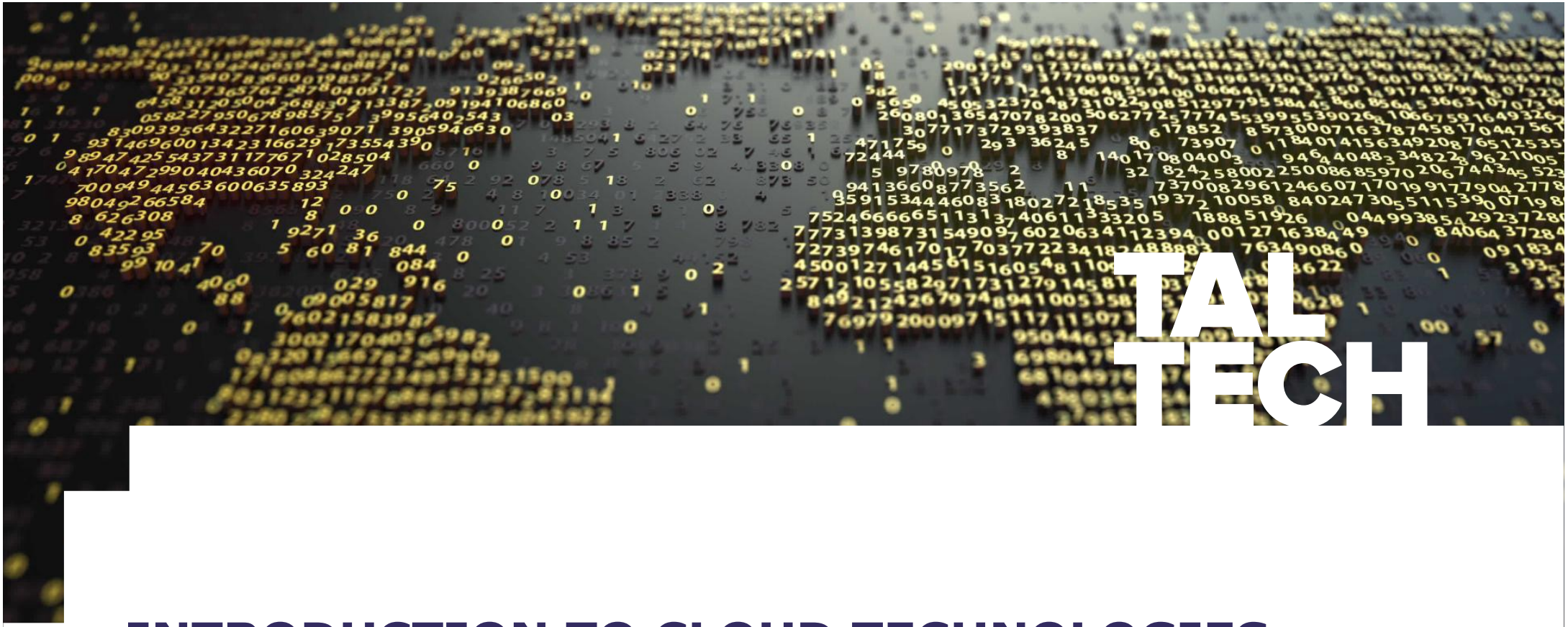
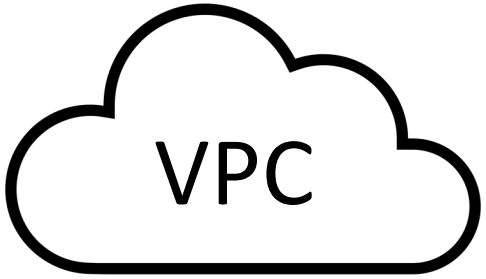


INTRODUCTION TO CLOUD TECHNOLOGIES (ICA0017)

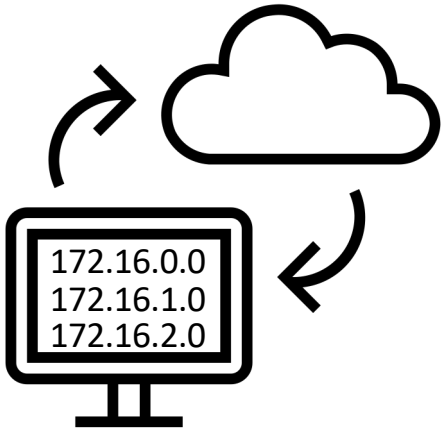
SIIM VENE

STEPS TO CREATE A VPC



Choosing an address range

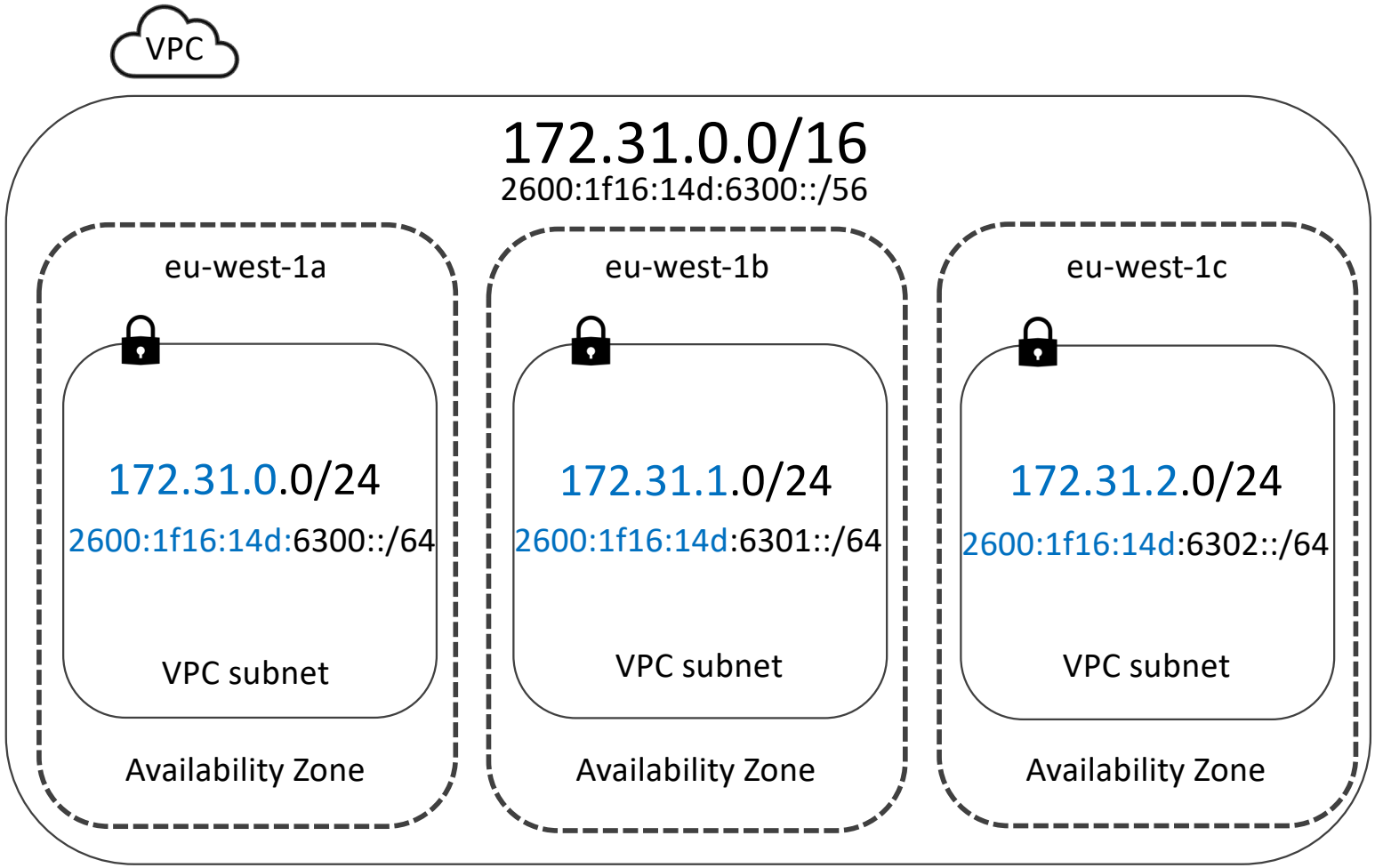
Create subnets in Availability Zones

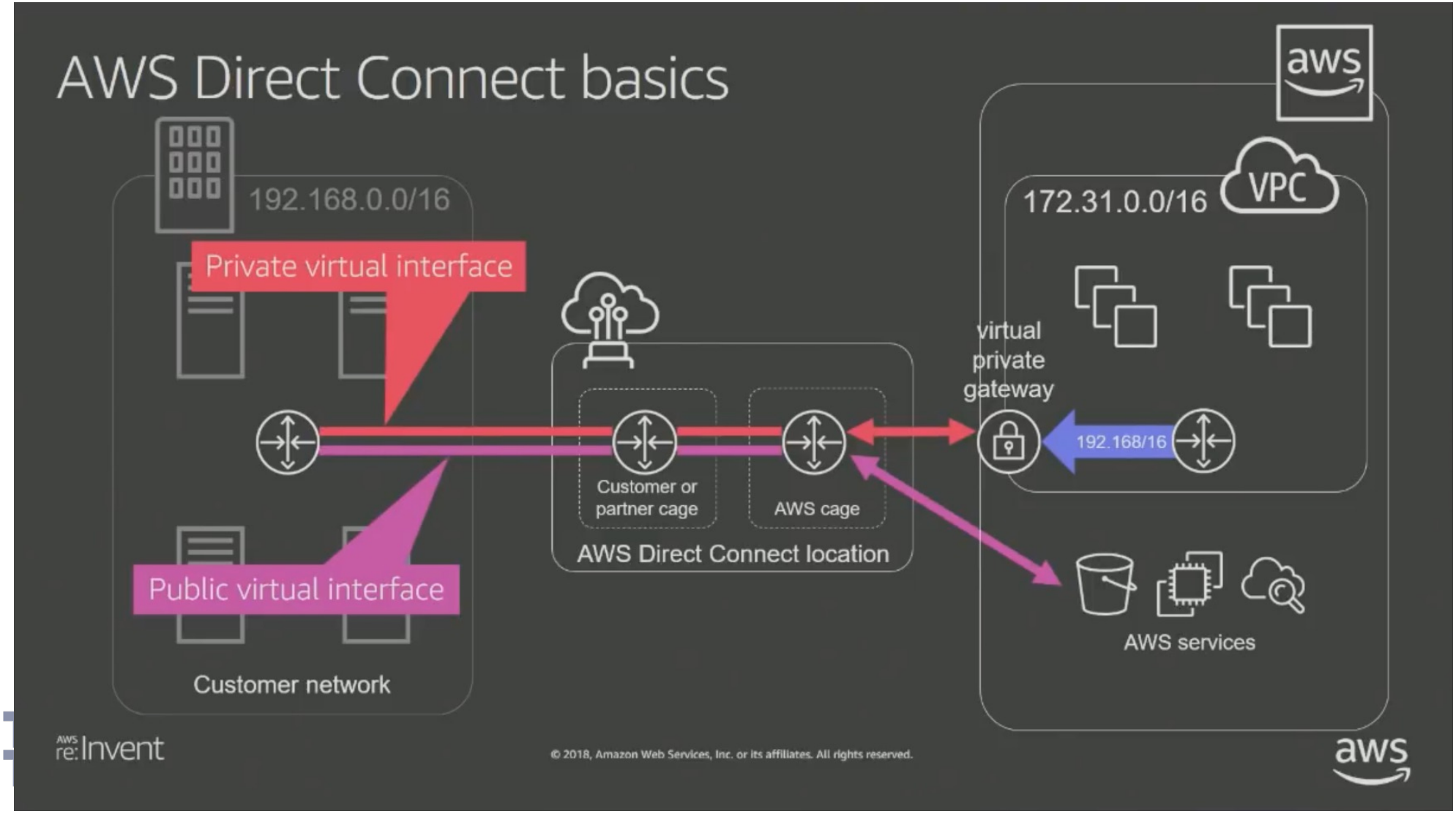


Create route to the internet

Authorize traffic to/from the VPC

TAL TECH


VPC
172.31.0.0/16
2600:1f16:14d:6300::/56
eu-west-1a
172.31.0.0/24
2600:1f16:14d:6300::/64
VPC subnet
Availability Zone
eu-west-1b
172.31.1.0/24
2600:1f16:14d:6301::/64
VPC subnet
Availability Zone
eu-west-1c
172.31.2.0/24
2600:1f16:14d:6302::/64
VPC subnet
Availability Zone


AWS Direct Connect basics
192.168.0.0/16
Private virtual interface
Public virtual interface
Customer network
Customer or partner cage
AWS cage
AWS Direct Connect location
virtual private gateway
172.31.0.0/16
VPC
192.168/16
AWS services
aws
re:Invent
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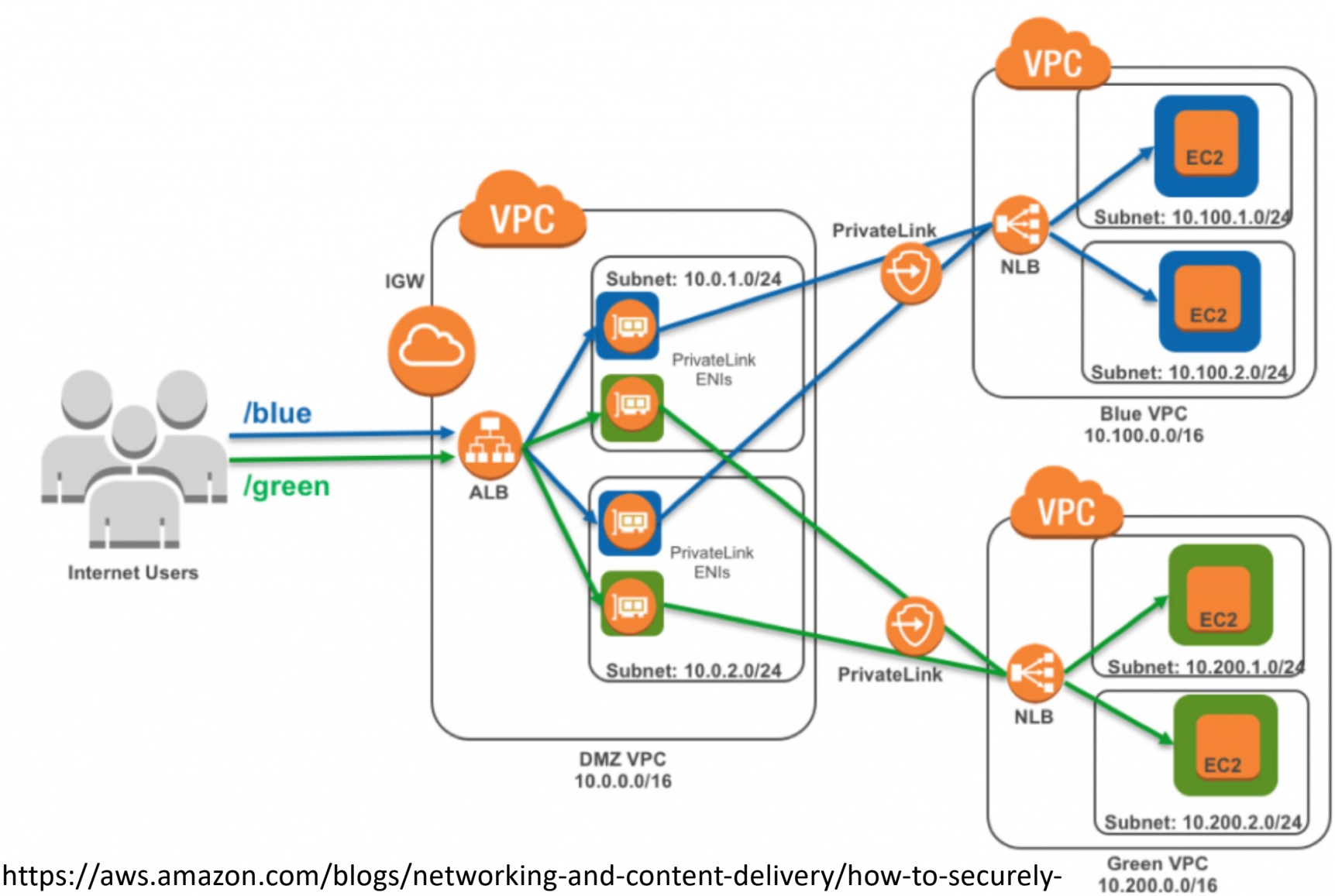
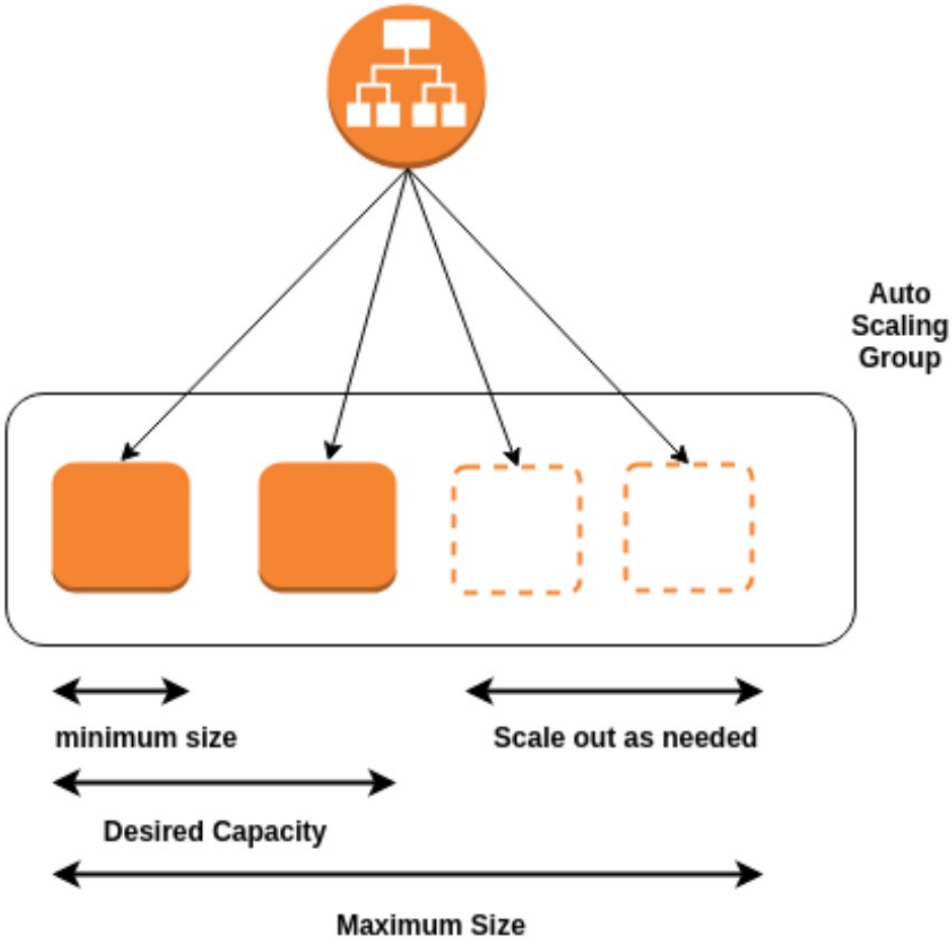


https://aws.amazon.com/blogs/networking-and-content-delivery/how-to-securely-publish-internet-applications-at-scale-using-application-load-balancer-and-aws-privatelink


Auto
Scaling
Group
minimum size
Scale out as needed
Desired Capacity
Maximum Size

TAL
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COURSE OVERVIEW

- 1. History, different service types (XaaS)
- 2. AWS, service portfolio, console, CLI, API
- 3. Compute services – EC2, serverless, containers
- 4. IAM services - users, groups, roles, governance
- 5. Storage services - S3, EBS, EFS, RDS
- 6. Network services - VPC, CDN
- **7. Logging, monitoring**
- 8. Automation, feasibility

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IT SECURITY MATTERS

Securing IT infrastructure is:

- Complex
- Expensive
- Time consuming – build/configure/maintain
- Difficult to track all the changes in IT environment
- Hard to do effectively

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WHAT HAPPENS IN CLOUD, ~~NEVER~~ STAYS IN CLOUD

How do you know:

- Who deployed a new EC2? Or deleted one?
- What application is unusable by the users?
- What needs to happen when 💩 hits the fan?
- Who needs to get notified?
- How to have an auditable trail of everything happening in your AWS account?

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MONITORING AND LOGGING

- Deep visibility into API calls
- Log aggregation and analytics
- Alert notifications
- Automated

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TWO MAIN SERVICES

AWS CloudTrail

Amazon
CloudWatch

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Amazon GuardDuty	Amazon Macie
Amazon Inspector	AWS Key Management Service (AWS KMS)
AWS Security Hub	AWS CloudHSM
Amazon Security Lake	AWS Certificate Manager
Amazon Detective	AWS Payment Cryptography
AWS Config	AWS Private Certificate Authority
Amazon CloudWatch	AWS Secrets Manager
AWS CloudTrail	AWS Artifact
AWS IoT Device Defender	AWS Audit Manager
AWS Elastic Disaster Recovery	

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AWS CLOUDTRAIL

Web service that records AWS API calls for your account and delivers log files for you

- Management evenets
- Data events
- Insight events

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MANAGEMENT EVENTS

Management events provide information about management operations that are performed on resources in your AWS account. These are also known as *control plane operations*.

- Configuring security (for example, AWS Identity and Access Management AttachRolePolicy API operations).
- Registering devices (for example, Amazon EC2 CreateDefaultVpc API operations).
- Configuring rules for routing data (for example, Amazon EC2CreateSubnet API operations).
- Setting up logging (for example, AWS CloudTrail CreateTrailAPI operations).

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CloudTrail > Event history

Event history (50+) Info

Download events | Create Athena table

Event history shows you the last 90 days of management events.

Lookup attributes

Resource type | AWS::EC2::Instance | 2024-03-11T00:00:00+02:00 — 2024-03-12T23:59:59+02:00

1 2 ...

Event name	Event time	User name	Resource type	Resource name
StopInstances	March 12, 2024, ...	stop_instance_afte...	AWS::EC2::Instance	i-04e484e166adf9b20
RunInstances	March 12, 2024, ...	emkarh@taltech.ee	AWS::EC2::VPC, AWS::E...	vpc-0864ffadf655558..
StopInstances	March 12, 2024, ...	tikund@taltech.ee	AWS::EC2::Instance, A...	i-0f80943513b484a17.
RunInstances	March 12, 2024, ...	tikund@taltech.ee	AWS::EC2::VPC, AWS::E...	vpc-0864ffadf655558..
SendSSHPublicKey	March 12, 2024, ...	tikund@taltech.ee	AWS::EC2::Instance	i-0efdc1575d7ccc6a7
SendSSHPublicKey	March 12, 2024, ...	tikund@taltech.ee	AWS::EC2::Instance	i-0f80943513b484a17
SendSSHPublicKey	March 12, 2024, ...	tikund@taltech.ee	AWS::EC2::Instance	i-0f80943513b484a17
RunInstances	March 12, 2024, ...	tikund@taltech.ee	AWS::EC2::VPC, AWS::E...	vpc-0864ffadf655558..

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DATA EVENTS

Data events provide information about the resource operations performed on or in a resource. These are also known as *data plane operations*. Data events are often high-volume activities.

- Amazon S3 object-level API activity on buckets and objects in buckets. (for example GetObject, DeleteObject, and PutObject API operations)
- AWS Lambda function execution activity (the Invoke API).
- CloudTrail PutAuditEvents activity on a CloudTrail Lake channel that is used to log events from outside AWS.
- Amazon SNS Publish and PublishBatch API operations on topics.

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INSIGHT EVENTS

CloudTrail Insights events capture unusual API call rate or error rate activity in your AWS account by analyzing CloudTrail management activity.

- Your account typically logs no more than **20** Amazon S3 deleteBucket API calls per minute, but your account starts to log an average of **100** deleteBucket API calls per minute. An Insights event is logged at the start of the unusual activity, and another Insights event is logged to mark the end of the unusual activity.
- Your account typically logs **20** calls per minute to the Amazon EC2AuthorizeSecurityGroupIngress API, but your account starts to log **zero** calls to AuthorizeSecurityGroupIngress. An Insights event is logged at the start of the unusual activity, and ten minutes later, when the unusual activity ends, another Insights event is logged to mark the end of the unusual activity.

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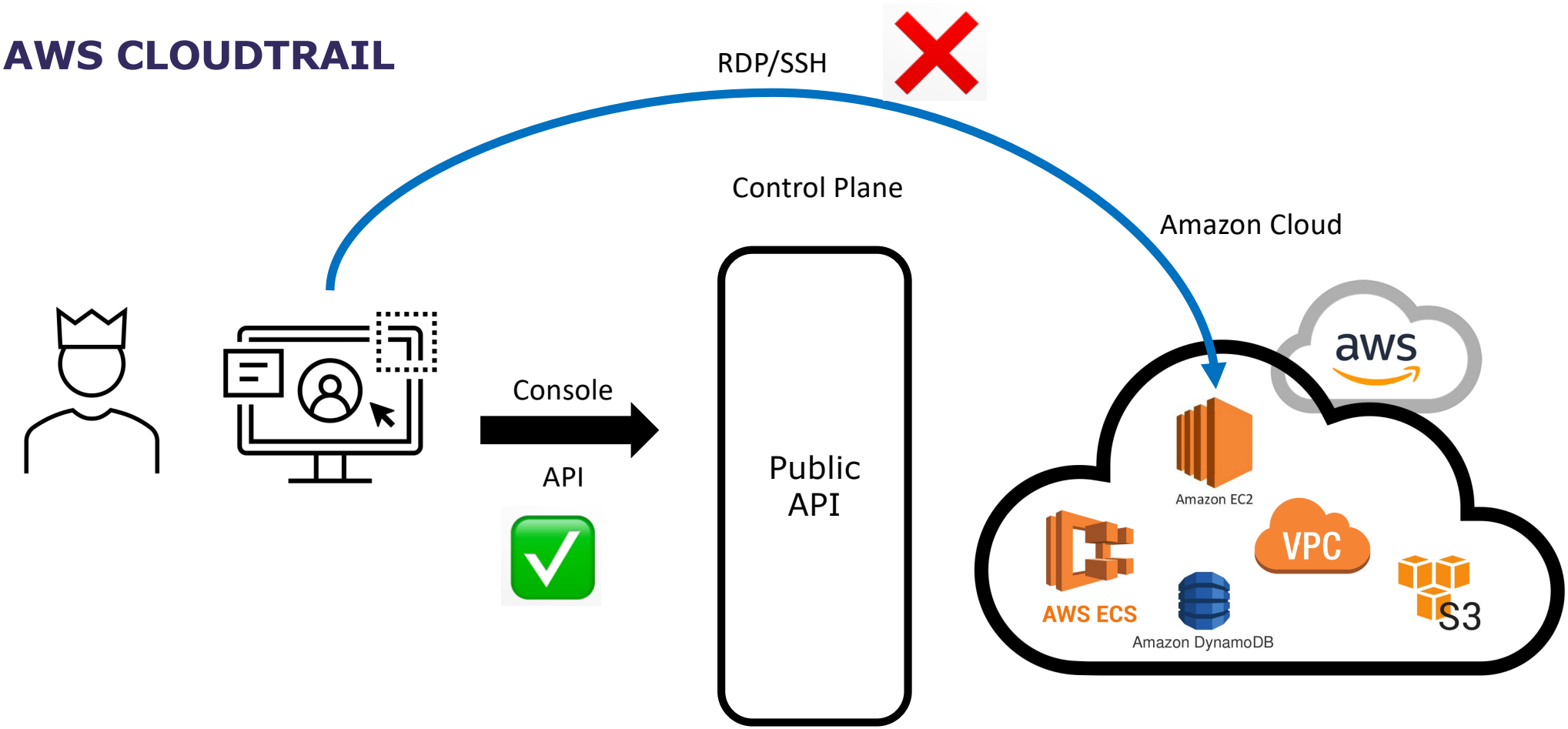
AWS CLOUDTRAIL



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AWS CLOUDTRAIL

Enables:

- After-the-fact incident investigation
- Near-realtime intrusion detection
- Industry and regulatory compliance
- Enabled by default for 7 days when you create an account

Provides:

- Logs API call details (for supported services)

https://docs.aws.amazon.com/awscloudtrail/latest/userguide/cloudtrail-aws-service-specific-topics.html

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WHAT IS LOGGED?

- Metadata of API calls
- Identity of API caller
- Time of the API call
- Source IP address of the API caller
- Request parameters
- Response elements retuned by the service

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WHERE IS IT LOGGED?

- Event logs are sent to an S3 bucket
- You manage the retention in S3!
- Delivered every 5 minutes up to 15-min delay
- Notifications available about the actions
- Can be aggregated across regions
- Can be aggregated across accounts

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CLOUDTRAIL LOG VALIDATION

The integrity of the logs is validated in addition:

- Was the log modified or deleted?
- Valdated by hashing every log file with SHA-256

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SECURING THE LOGS

If CloudTrail logs contains metadata, not application data, why bother securing them?

Logs may contain PII data such as usernames and team/group membership. Also can contain detailed configuration information

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SECURING THE LOGS

- Use IAM policies and S3 bucket policies to restrict access to S3 bucket containing the logs
- Use SSE-S3 or SSE-KMS to encrypt the logs
- Configure SNS notifications and log file validation
- Restrict S3 delete access

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ACTING UPON LOGS

- CloudTrail itself does nothing with logs
- You can review logs in the CloudTrail console
- Utilizes CloudWatch Events, to define workflows
- Detect unusual activity by enabling CloudTrail insights

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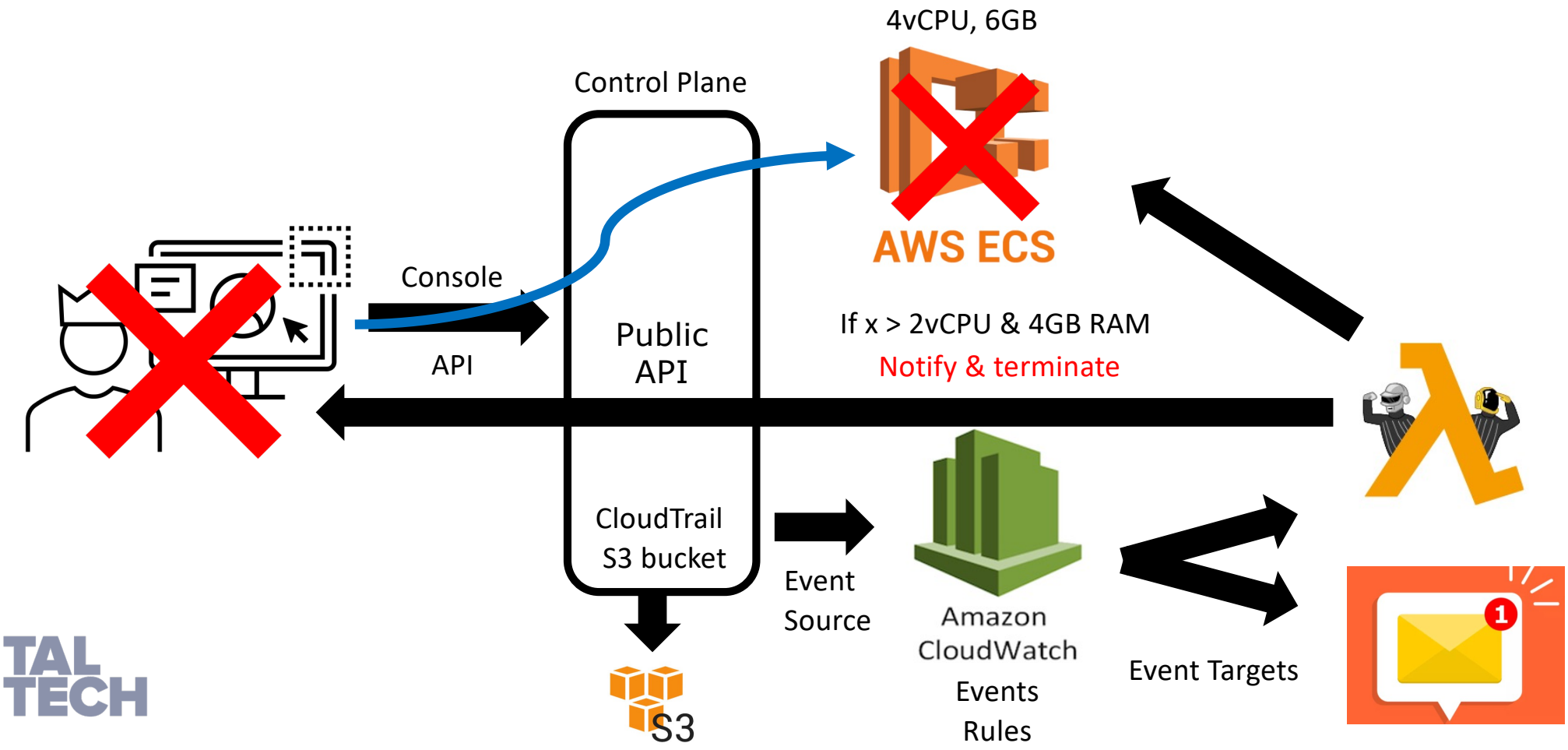
AWS CLOUD TRAIL



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AWS CLOUDWATCH

A monitoring service for AWS cloud resources and the applications **you run on AWS**

Key components:

- CloudWatch
- CloudWatch Logs
- CloudWatch Events

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AWS CLOUDWATCH

Enables:

- Resource utilization, operational performance monitoring
- Log aggredation and basic analysis

Provides:

- Real time monitoring within AWS for resources and application
- Hooks to event triggers

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CLOUDWATCH

- Real time
- Metrics
- Alarms
- Notifications
- Custom metrics

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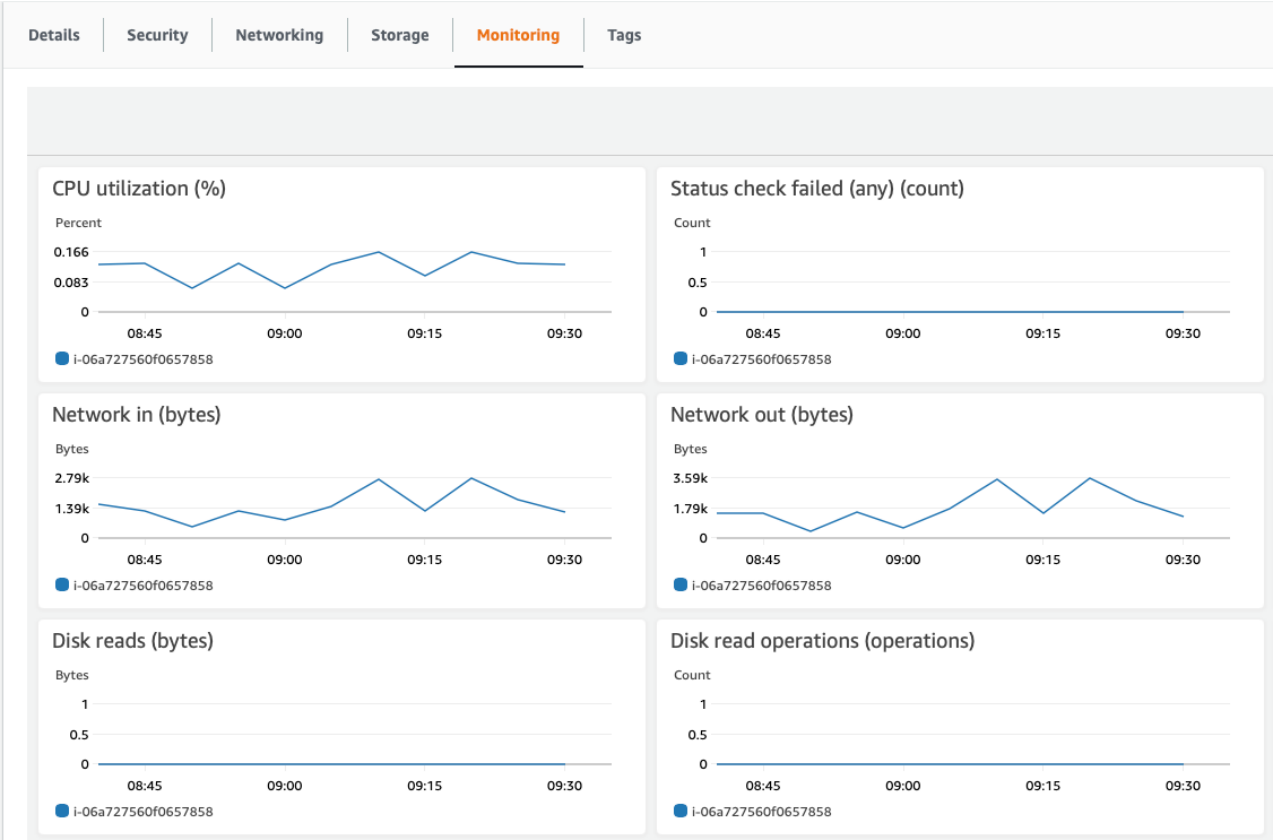
Details
Security
Networking
Storage
Monitoring
Tags
CPU utilization (%)
Percent
0.166
0.083
0
08:45
09:00
09:15
09:30
i-06a727560f0657858
Status check failed (any) (count)
Count
1
0.5
0
08:45
09:00
09:15
09:30
i-06a727560f0657858
Network in (bytes)
Bytes
2.79k
1.39k
0
08:45
09:00
09:15
09:30
i-06a727560f0657858
Network out (bytes)
Bytes
3.59k
1.79k
0
08:45
09:00
09:15
09:30
i-06a727560f0657858
Disk reads (bytes)
Bytes
1
0.5
0
08:45
09:00
09:15
09:30
i-06a727560f0657858
Disk read operations (operations)
Count
1
0.5
0
08:45
09:00
09:15
09:30
i-06a727560f0657858

CLOUDWATCH LOGS

- Pushed from any AWS service (including CloudTrail)
- Pushed from your application/system
- Can also be utilized for on-premises/non-AWS workloads
- Metrics from log entry matches
- Stored infefinitely (not your S3)

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CLOUDWATCH EVENTS

Near real time stream on system events

- Events:
- AWS resources state change
- AWS Cloudtrail (API calls)
- Custom events (code)
- Sheduled

- Rules – actinoable items that match events with targets
- Targets – AWS lambda functions, SNS topics etc

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NOTIFICATIONS

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SNS (SIMPLE NOTIFICATIONS SERVICE)

SNS is a managed service that provides message delivery from publishers to subscribers (also known as *producers* and *consumers*)

- Publishers communicate asynchronously with subscribers by sending messages to a *topic*, which is a logical access point and communication channel.
- Clients can subscribe to the SNS topic and receive published messages using a supported endpoint type, such as Amazon Data Firehose, Amazon SQS, AWS Lambda, HTTP, email, mobile push notifications, and mobile text messages (SMS).

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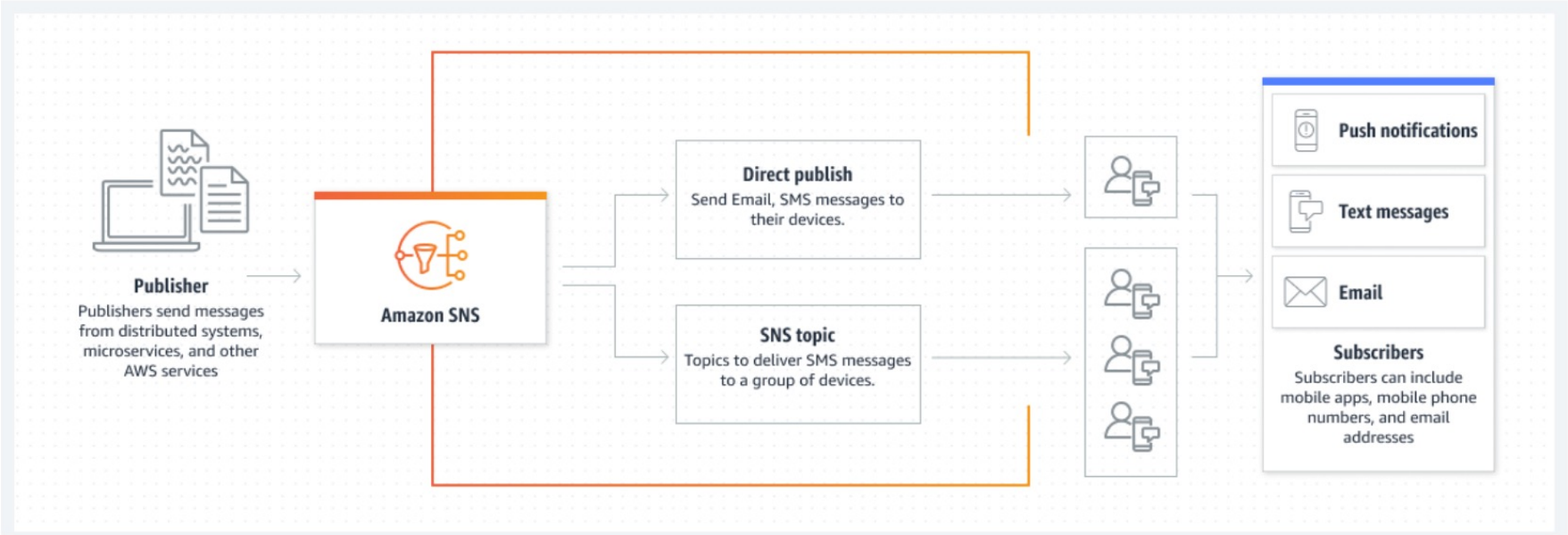
Publisher
Publishers send messages from distributed systems, microservices, and other AWS services
Amazon SNS
Direct publish
Send Email, SMS messages to their devices.
SNS topic
Topics to deliver SMS messages to a group of devices.
Push notifications
Text messages
Email
Subscribers
Subscribers can include mobile apps, mobile phone numbers, and email addresses

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